

» surface. Associated with geometry and optics, painting was part of the mainstream of intellectual development in the Renaissance.

Overtaking the ancients

The striving for an exact depiction of the world in art was related to a wider trend toward the close observation of nature, rather than reliance on received wisdom. This produced clear advances on the knowledge inherited from the ancients. For instance, the dissection of corpses by the Brussels-born anatomist Andreas Vesalius at the University of Padua allowed the long-accepted views of Galen, the revered medical authority of antiquity, to be superseded. The geographical notions of the ancient Greek Ptolemy were corrected thanks to the efforts of European explorers and mapmakers. Nicolaus Copernicus challenged the current Christian view of the universe by asserting that the Sun, not Earth, was at the center of the solar system (see p.266).

Craftsmanship

Underpinning the achievements of the celebrated geniuses of the Renaissance was a widespread culture of practical ingenuity and skilled craftsmanship. Fine craft objects, such as the elaborate suits of armor made by the metalworkers of Nuremberg and Augsburg, or the products of Venetian or Florentine goldsmiths, were collected as avidly as any painting or sculpture. More practically, it was during this time that gunpowder weapons came into general use in European warfare

Timepiece

The first clocks marked with minutes appeared in the 1570s (see below), but truly accurate timekeeping was not possible until pendulum clocks were introduced in the 17th century.



A dagger carries an abbreviated Latin inscription, AET. SVAE 29. This gives the age of the subject, France's ambassador to England Jean de Dinteville, as 29.

Jean de Dinteville is depicted with his friend, French envoy Georges de Selve. The objects between them suggest their wealth and learning.

The celestial globe, showing the constellations, was the latest model, made by German astronomer Johannes Schöner. Its presence echoes the terrestrial globe below.

The scientific instruments are devices for navigation and time measurement. They are resting on a valuable Turkish rug, imported from the Ottoman Empire.



The Ambassadors

Painted by German artist Hans Holbein at the court of English king Henry VIII in 1533, this painting shows the Renaissance's fascination with scientific instruments as well as its taste for hidden meanings and visual tricks.

In the foreground is a distorted representation of a skull, an obvious symbol of mortality. The skull appears correctly if viewed from a precise low point to the side of the painting.

The open book is a Lutheran hymnal, open at the hymns "Come Holy Ghost" and "Man Wilt Thou Live Blessedly." It probably hints at the religious conflicts of the Reformation.

(see pp.272–73). Mechanical clocks, which had first started to appear in the towers of Italian cities in the 14th century, became more widespread. Navigational instruments such as magnetic compasses were essential tools of expanding seaborne trade.

The printed word

By far the most influential product of European enterprise, however, was the printing press. Fresh thinking became readily available through the relative speed with which

new printed books could be produced and distributed. Literature in the vernacular—everyday languages such as English, Italian, and French, rather than scholarly Latin—received an enormous boost. Humanist scholars were able to circulate accurate, standardized versions of major texts. The most important of these was the Bible. Providing people with direct access to what was regarded as the word of God was a prime cause of the great religious upheaval of the Reformation (see pp.256–57).

Political turmoil

The cultural developments of the Renaissance took place within a context of acute political and religious conflicts. The Italian city-states that were such dynamos of creativity were also the site of vicious power struggles. In Florence the dominant Medici family was twice driven into exile by popular uprisings, in 1494 and 1527, and twice restored to power. Civil conflicts were complicated by the intervention of the armies of the kings of France and the Habsburg emperors,